

Option A: Immunology and Disease

7. Insect bites can cause infection of the deeper layers of the skin. This infection is called cellulitis and is caused by bacteria such as *Streptococcus* and *Staphylococcus* that usually live harmlessly on the surface of the skin. These bacteria are introduced into the wound when the bite is scratched. Flucloxacillin is a bactericidal antibiotic, similar to penicillin, which is used to treat insect bite infections.

(a) (i) State what is meant by the term bactericidal. [1]

(ii) *Streptococcus* and *Staphylococcus* are types of Gram-positive bacteria. Suggest how and why flucloxacillin has a bactericidal effect on these types of bacteria. [4]

(iii) Flucloxacillin has become ineffective against some species of *Staphylococcus*. Suggest why. [3]

(iv) Suggest how infection following the insect bite could be prevented. [1]



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- (b) Malaria is a disease spread by insect bites from some mosquitoes. They carry the protoctistan parasite *Plasmodium*, which causes the disease. Malaria is endemic in some sub-tropical areas, can become epidemic during wet seasons and could also be regarded as pandemic. **Image 7.1** shows the distribution of malaria.

Image 7.1

Malaria cases (per 100 000) by country



- (i) Describe the difference between the terms endemic and epidemic. Suggest why malaria could be considered pandemic. [2]

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- (ii) State why there is no current effective vaccine against *Plasmodium*. [2]

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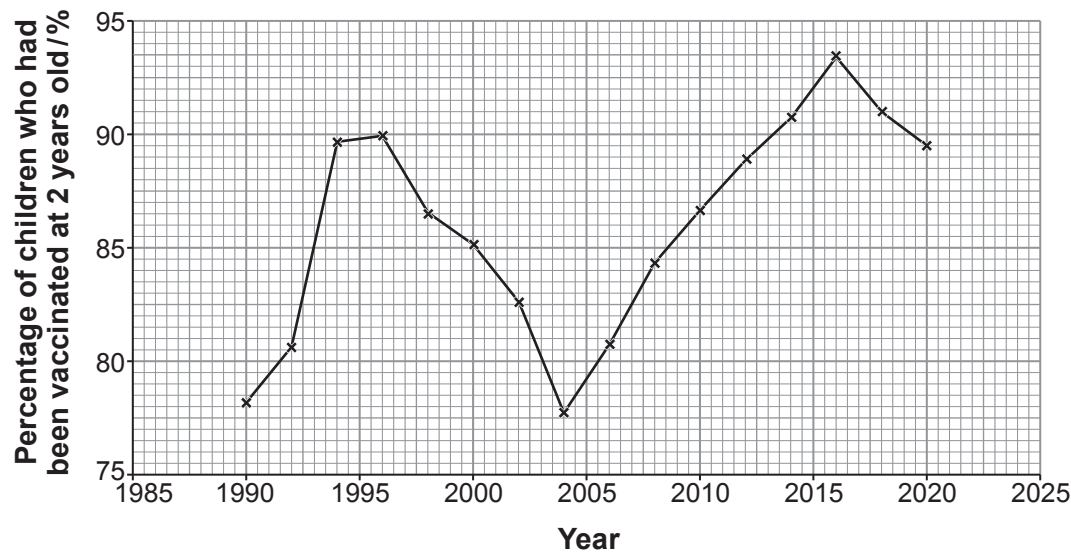
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- (c) The aim of vaccination is to develop humoral and cell-mediated responses against a number of potentially fatal diseases. In 2019, the UK lost its World Health Organisation (WHO) measles-free status as the number of cases of measles was rising. The MMR vaccine provides immunity for measles, mumps and rubella. **Graph 7.2** shows the results of a study into the percentage uptake of the vaccine in children.

Graph 7.2



- (i) Conclude why the number of cases of measles increased between 2016 and 2020. [1]

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- (ii) In 2018 there were 700 000 children born. Calculate the number of children that had not been vaccinated by the time they were **two years old**. [2]

Number of children =



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(iii) The MMR vaccine contains antigens of the three pathogens. Explain why a child who is exposed to the measles virus after being vaccinated does not develop the measles disease. [3]

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The MMR vaccine was linked to autism developing in children in a report that has since been disproven. However, some people still associate this condition with the combined MMR vaccine.

(iv) Use **Graph 7.2** to suggest which year the report was published. [1]

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